Turbulence forecasting · validation

Model: yalamanchi_vit | checkpoint epoch: 100

Training: five-step rollout, normalized MSE + 1 × relative KE error², horizon 5

Forecast: 10 recursive steps; no ground-truth resets

Start: file 451, frame 0, t=121.722496

End: t=122.022331 | elapsed simulation time: 0.299835

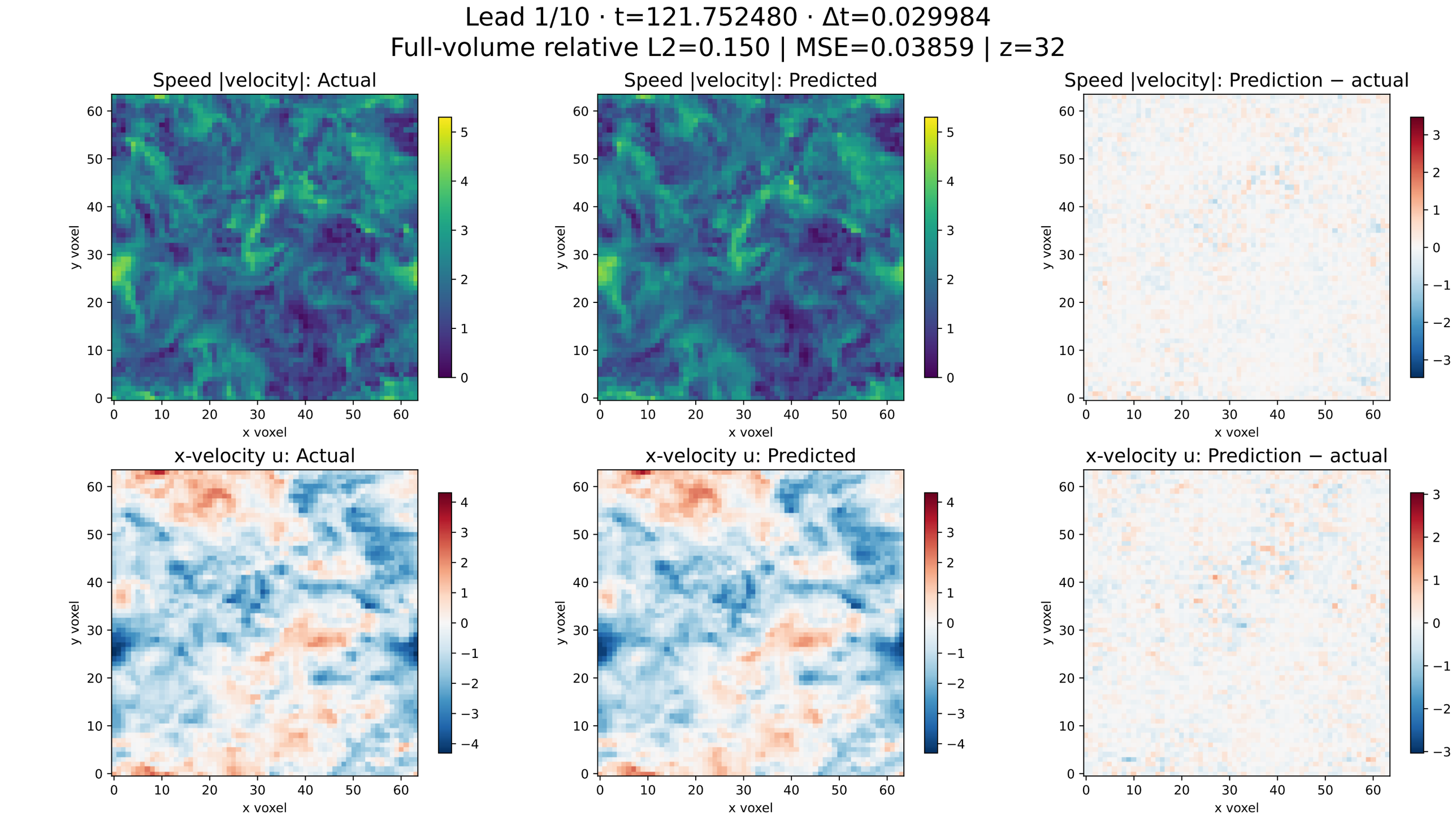
Filtered 64^3 velocity data | fixed z=32 slice

Fixed color scales across all leads; errors are prediction minus actual.

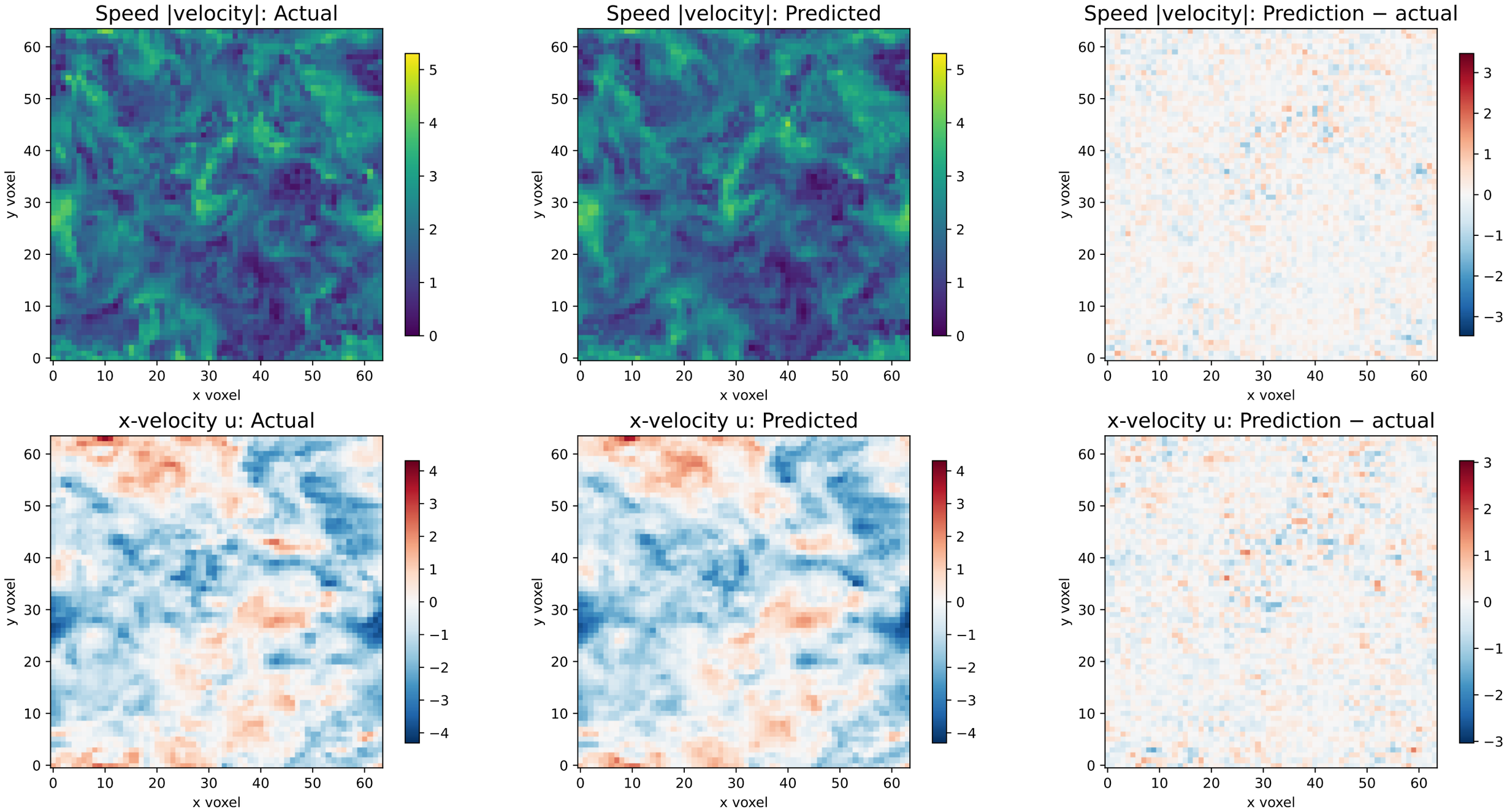
Field units are undocumented; metrics use original cached field values.

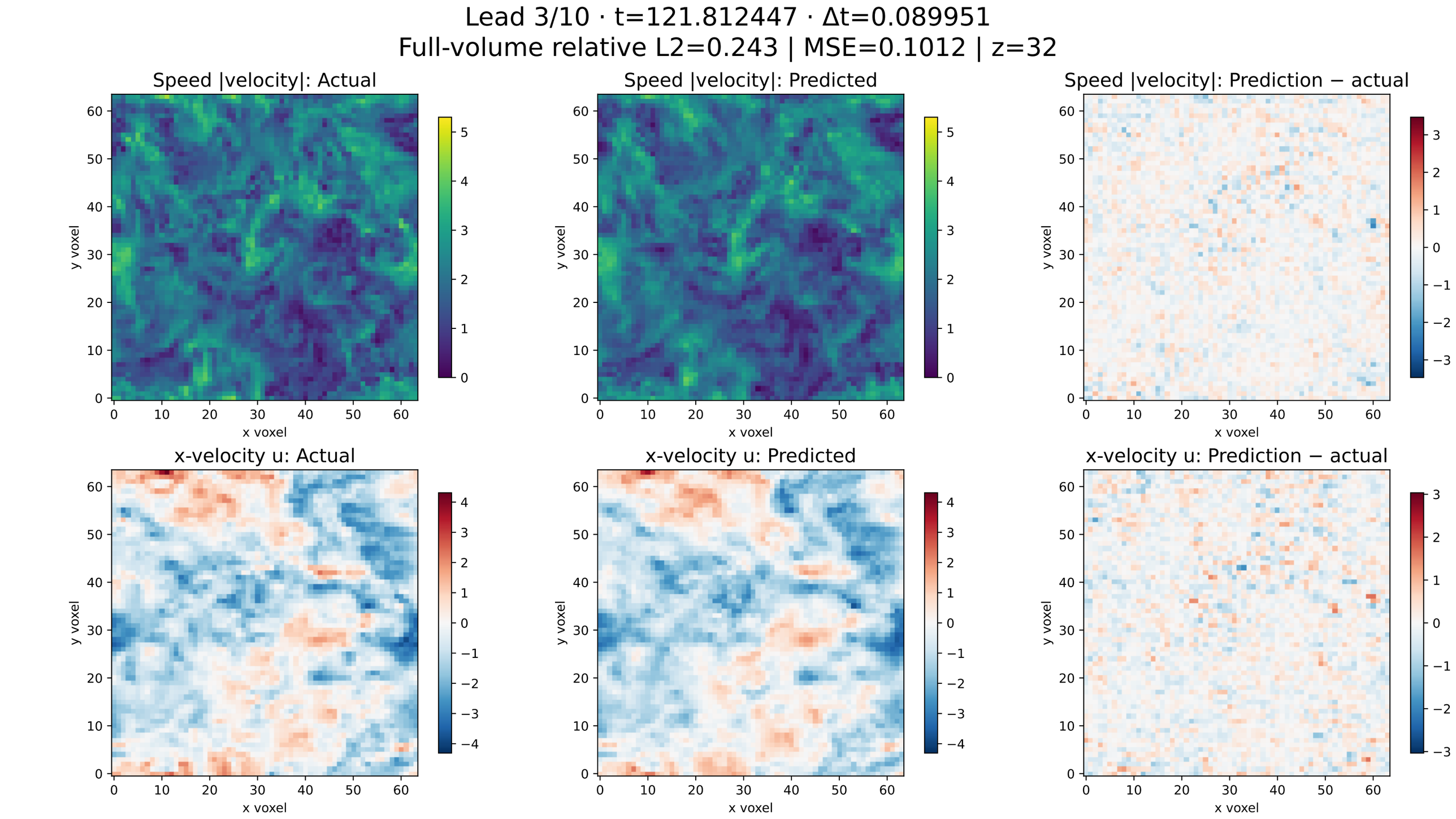
This trajectory is the first validation window, selected before evaluation.

Validation was used for checkpoint selection; these are not independent test results.

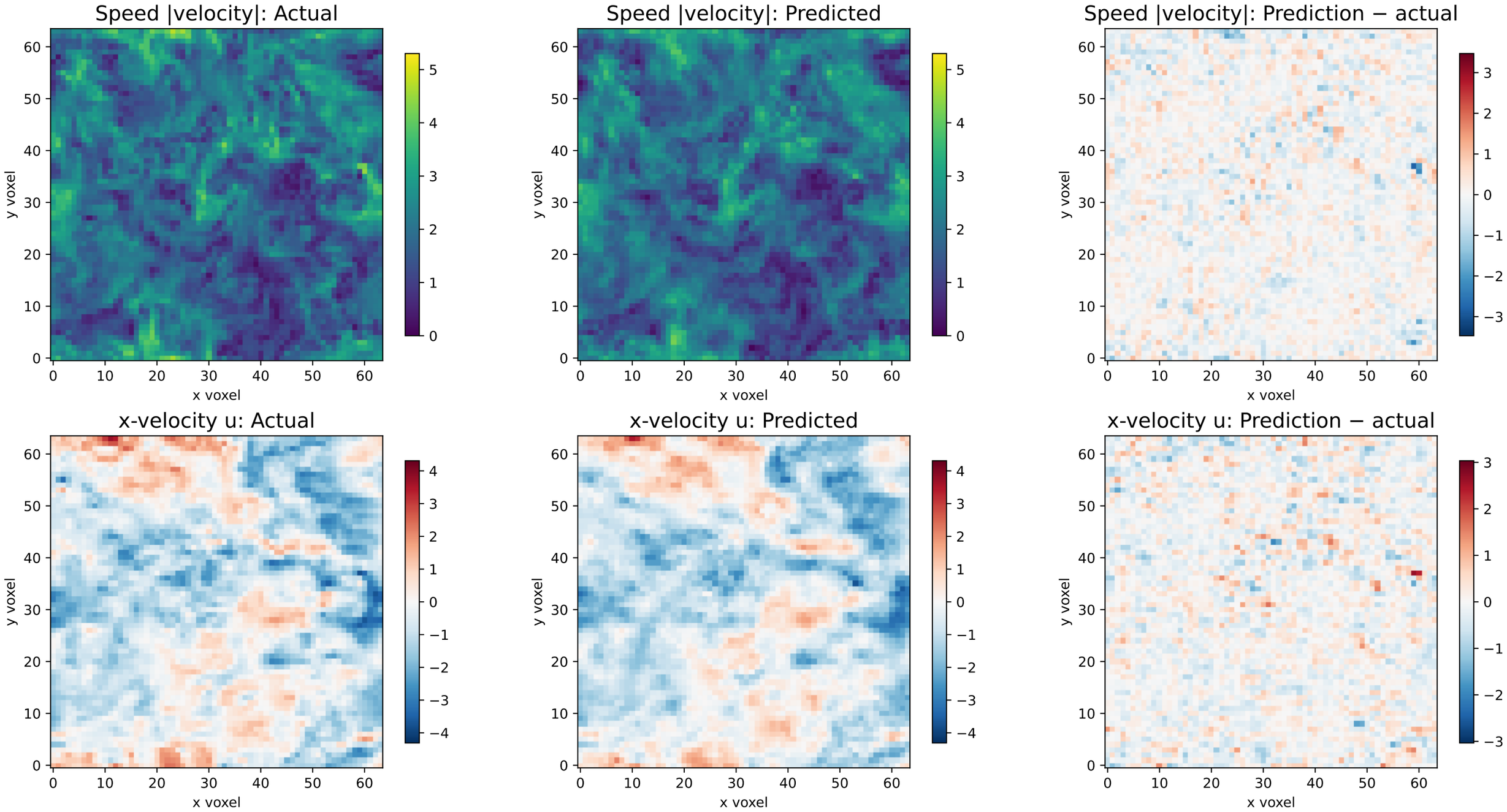

Lead 1/10 · t=121.752480 · Δt=0.029984
Full-volume relative L2=0.150 | MSE=0.03859 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel

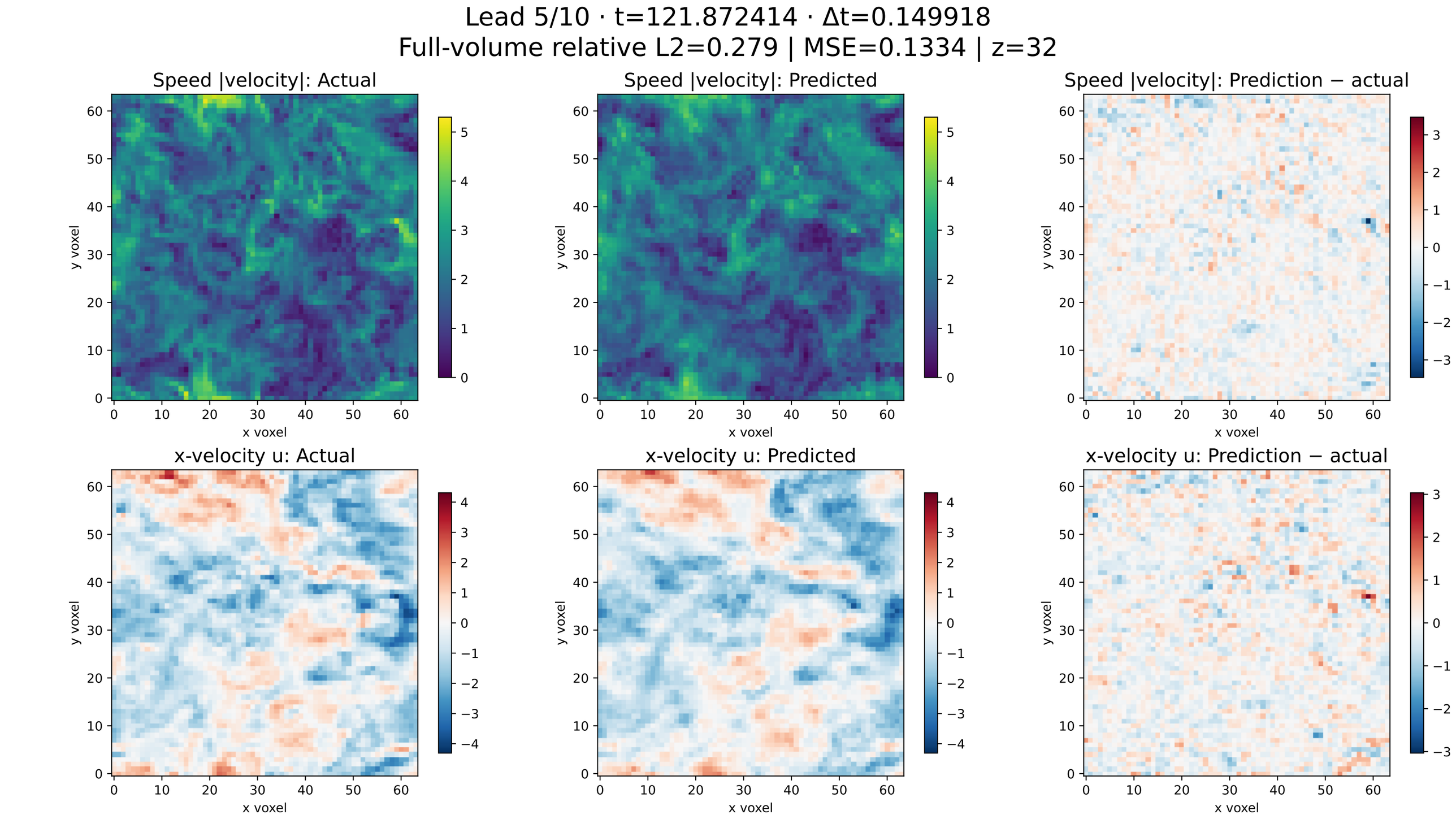
Lead 2/10 · t=121.782463 · Δt=0.059967
Full-volume relative L2=0.215 | MSE=0.07916 | z=32



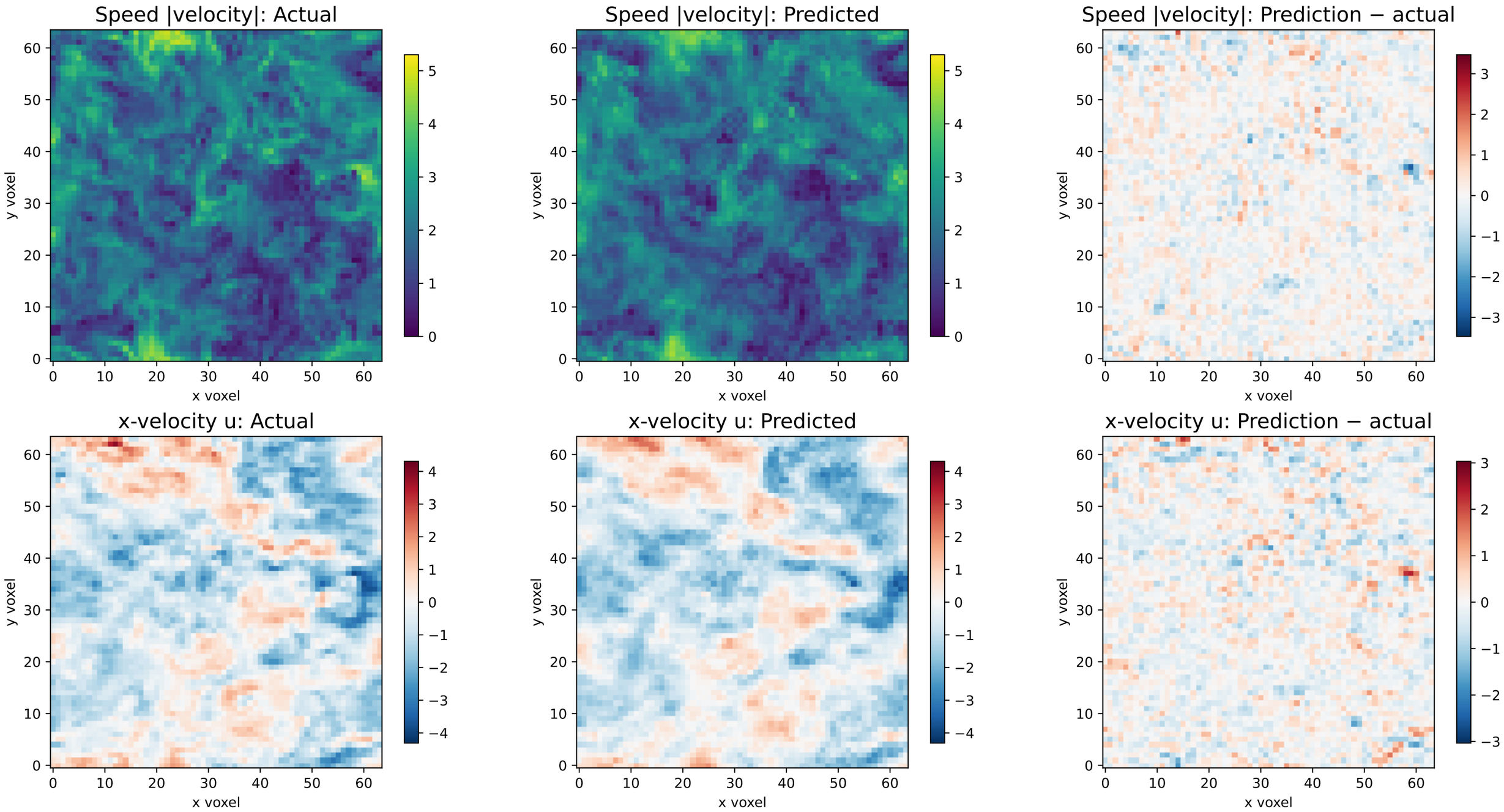

Lead 3/10 · t=121.812447 · Δt=0.089951
Full-volume relative L2=0.243 | MSE=0.1012 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel

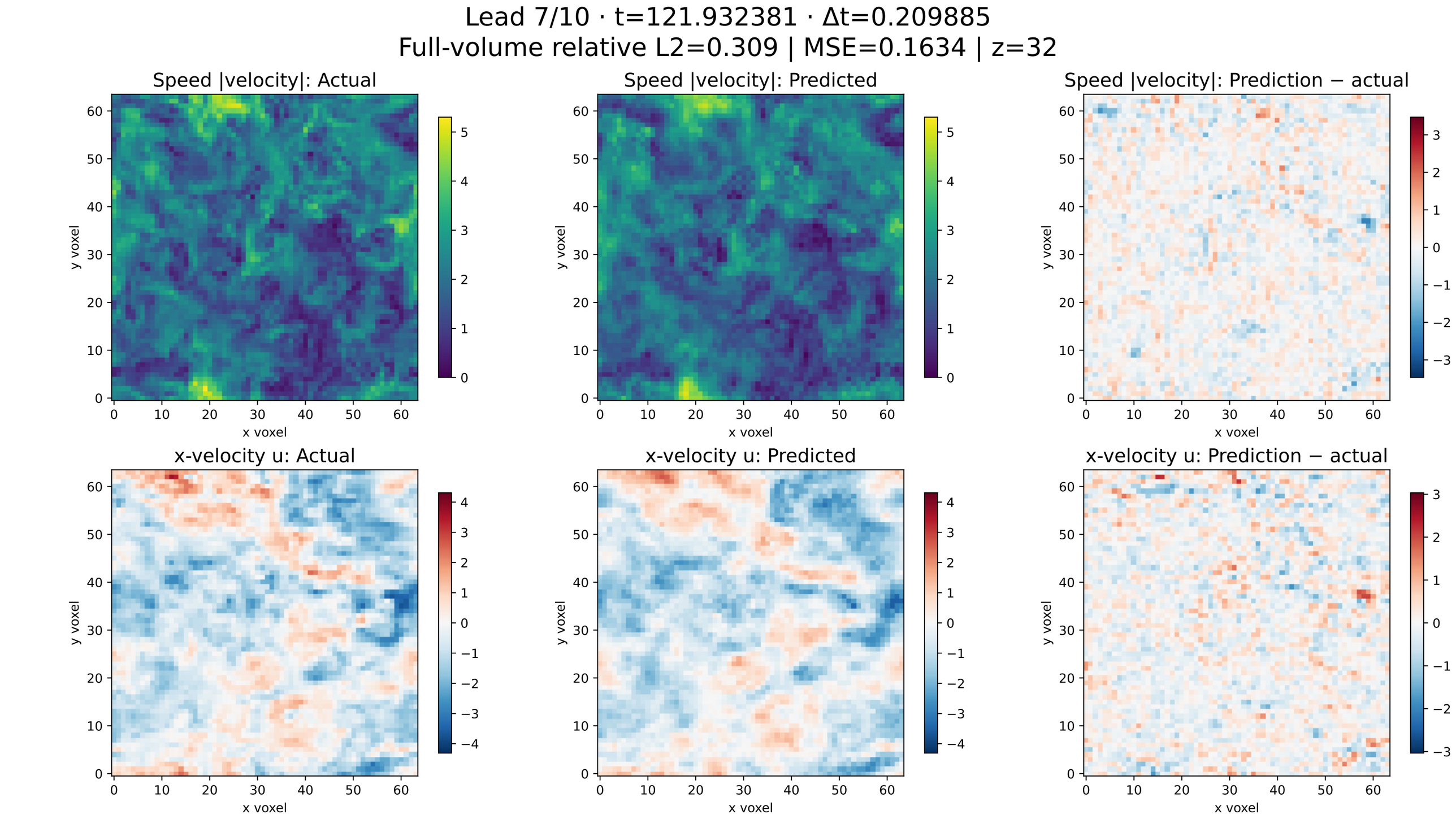
Lead 4/10 · t=121.842430 · Δt=0.119934
Full-volume relative L2=0.263 | MSE=0.118 | z=32

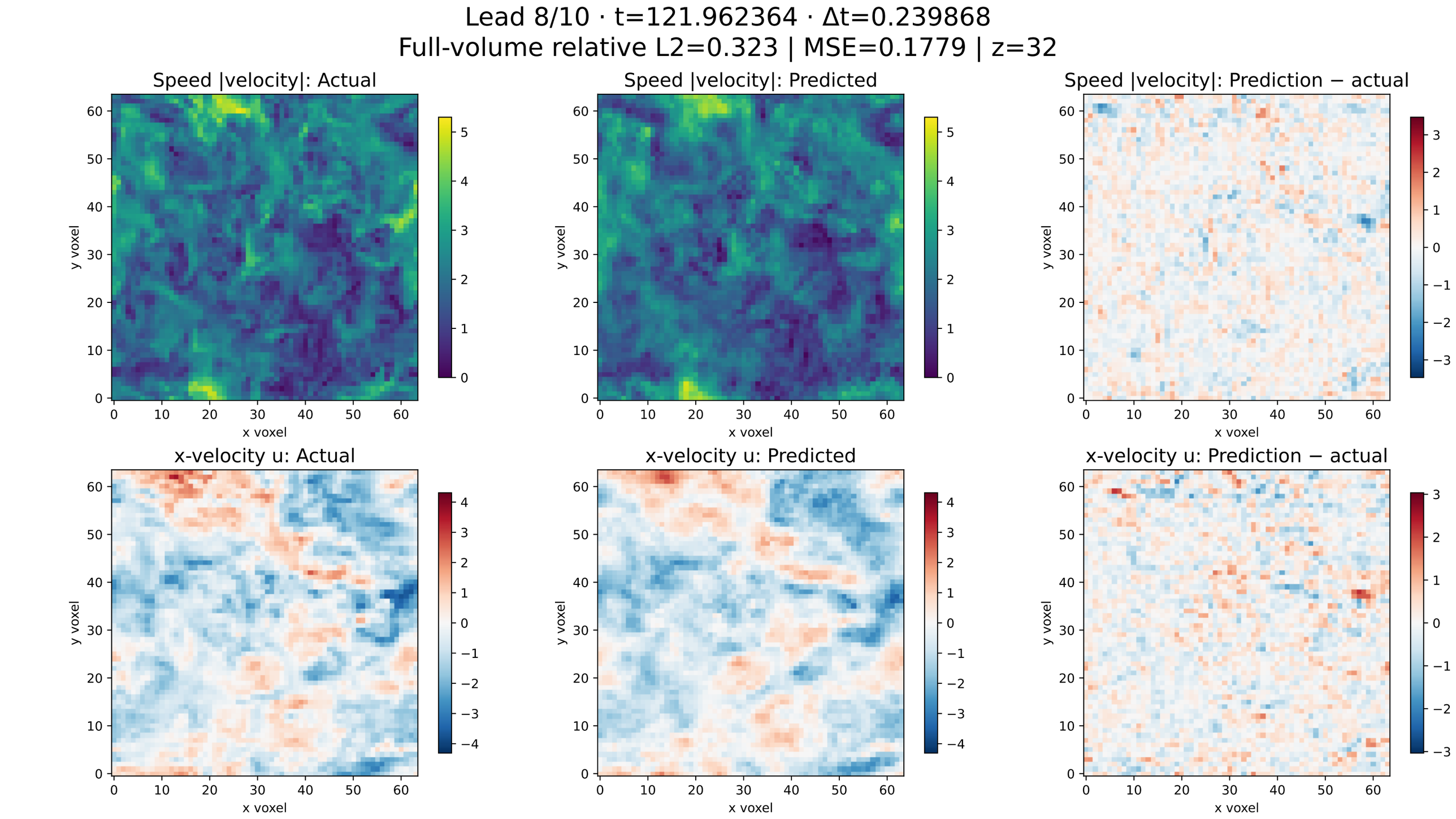


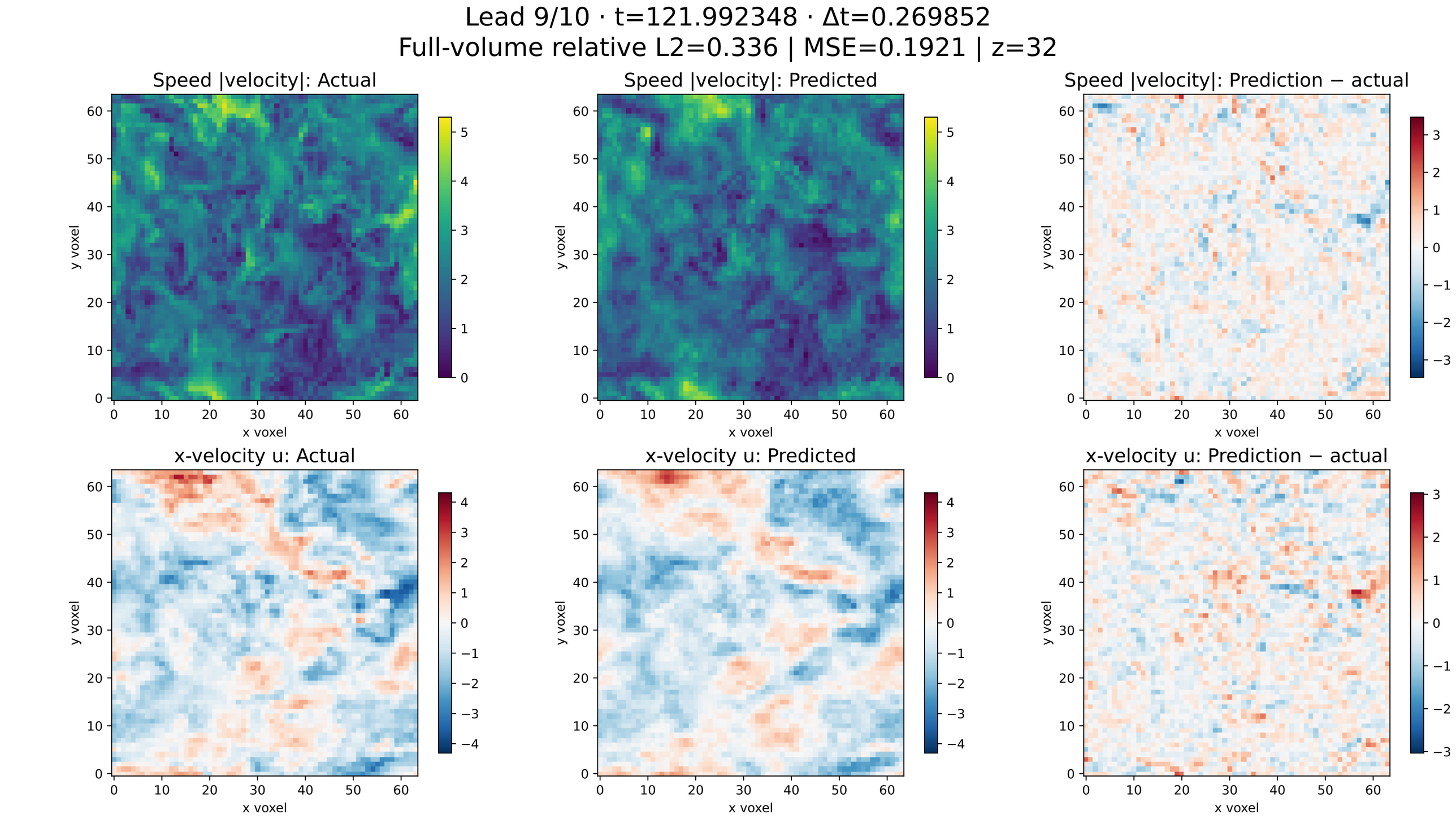

Lead 5/10 · t=121.872414 · Δt=0.149918
Full-volume relative L2=0.279 | MSE=0.1334 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel

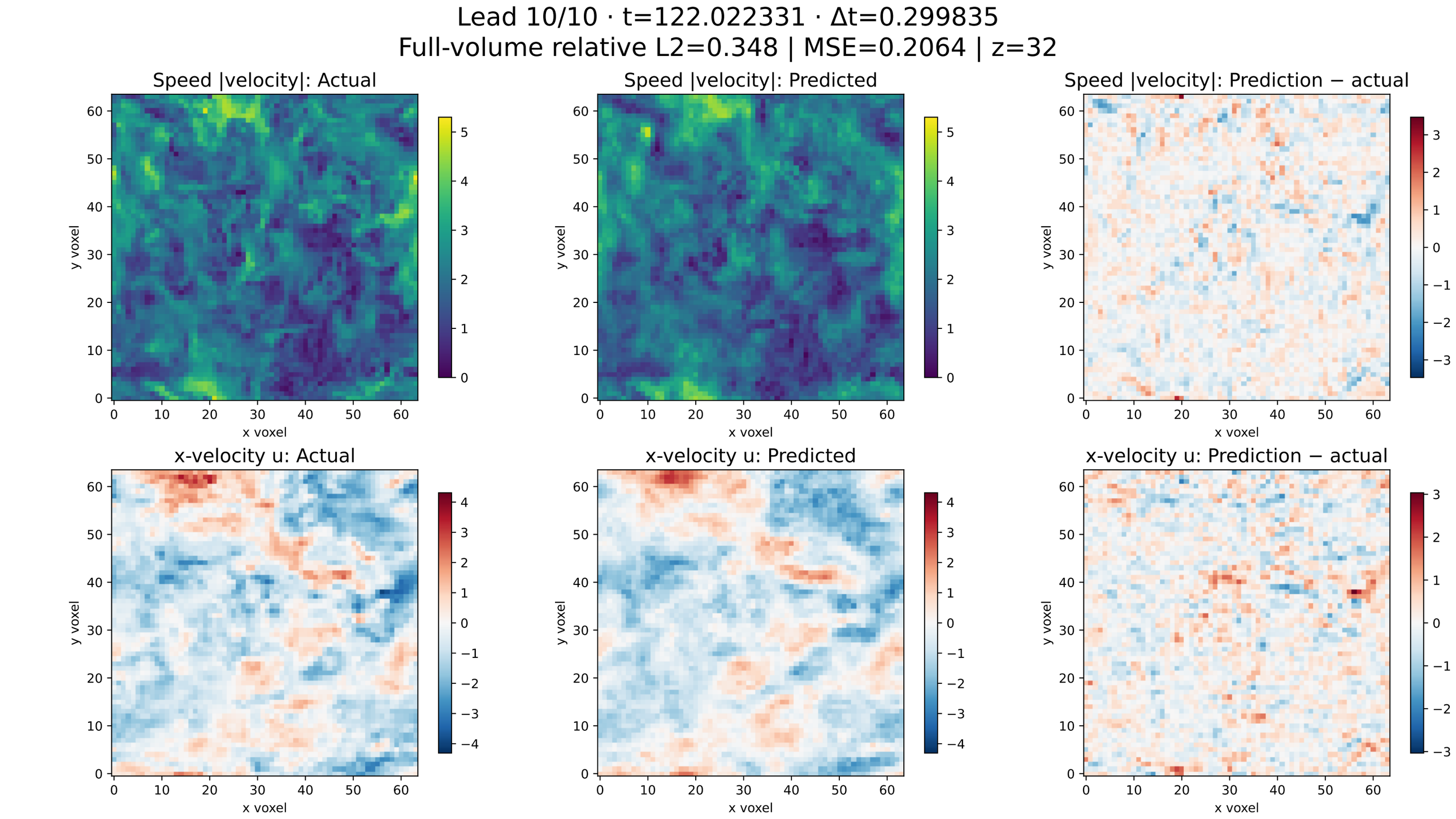
Lead 6/10 · t=121.902397 · Δt=0.179901
Full-volume relative L2=0.295 | MSE=0.1484 | z=32

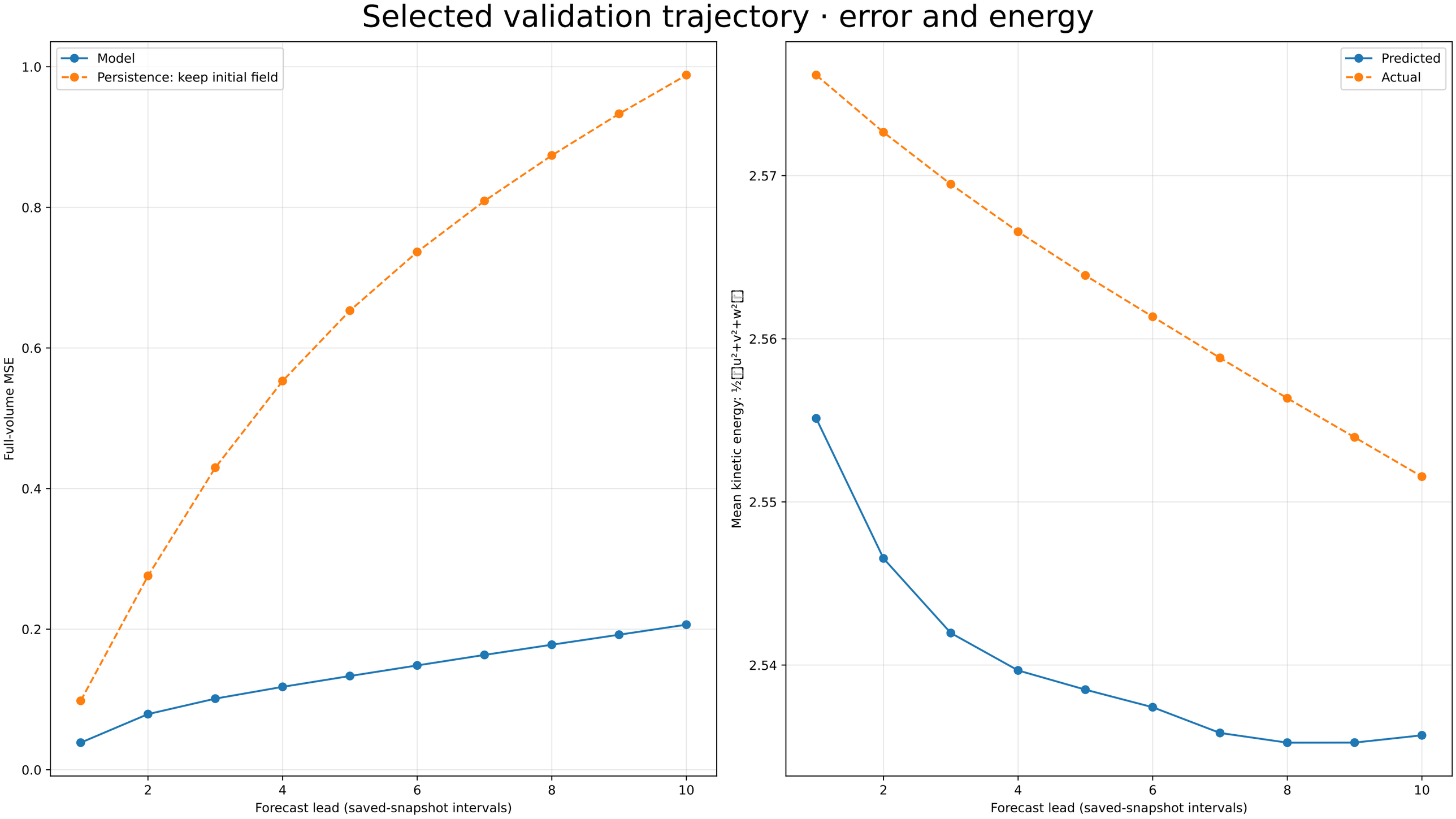



Lead 7/10 · t=121.932381 · Δt=0.209885
Full-volume relative L2=0.309 | MSE=0.1634 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel


Lead 8/10 · t=121.962364 · Δt=0.239868
Full-volume relative L2=0.323 | MSE=0.1779 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
y voxel
x voxel


Lead 9/10 · t=121.992348 · Δt=0.269852
Full-volume relative L2=0.336 | MSE=0.1921 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
y voxel
x voxel


Lead 10/10 · t=122.022331 · Δt=0.299835
Full-volume relative L2=0.348 | MSE=0.2064 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel


Selected validation trajectory · error and energy
Model
Persistence: keep initial field
Full-volume MSE
1.0
0.8
0.6
0.4
0.2
0.0
2
4
6
8
10
Forecast lead (saved-snapshot intervals)
Predicted
Actual
Mean kinetic energy: ½⟨u²+v²+w²⟩
2.57
2.56
2.55
2.54
2
4
6
8
10
Forecast lead (saved-snapshot intervals)